

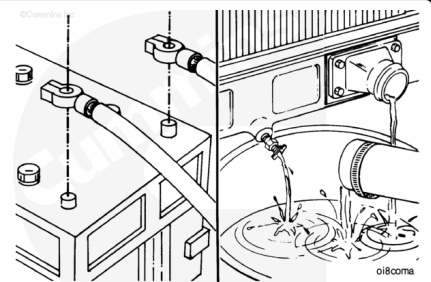
Trainings ▾

Remove

ISM Engines

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and connect the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

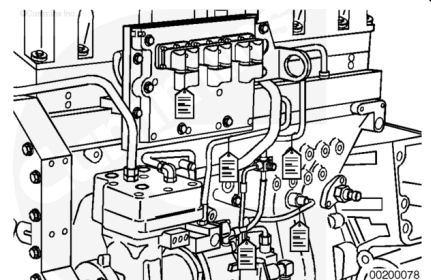
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

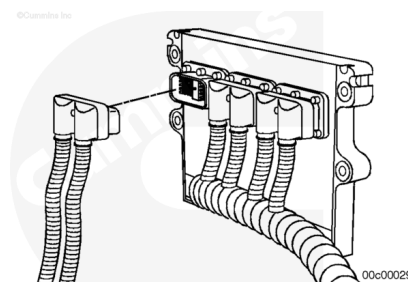
Disconnect the battery cables.

Drain the engine coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).

Place a tag on all hoses, lines, linkage, and electrical connections as they are removed, to identify their locations.



Disconnect the OEM wiring harness from the ECM. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/82/82-019-031.html) in the Troubleshooting and Repair Manual, Electronic Control System, ISM and QSM11 Engines, Bulletin 3666266.

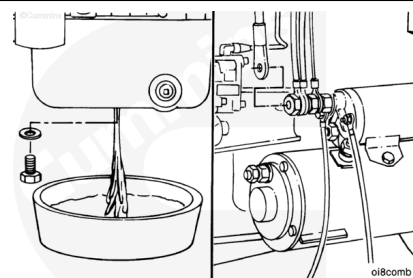


⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

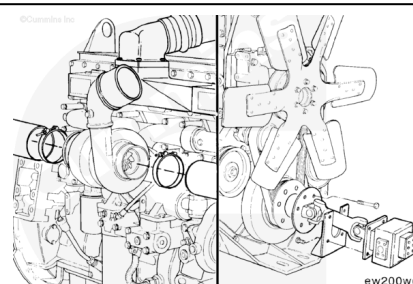


Drain the lubricating oil. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/101/101-007-025.html).

Disconnect the starter cable, engine ground straps, cab or chassis to engine hoses, tubing, electrical wires and hydraulic lines.

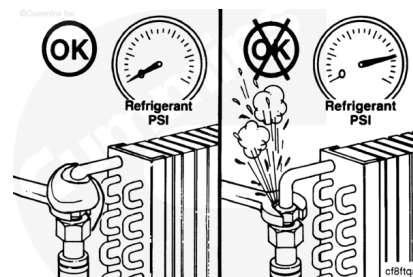
Disconnect the intake and exhaust air pipes.

Disconnect all chassis-mounted engine-driven accessories.



⚠ WARNING ⚠

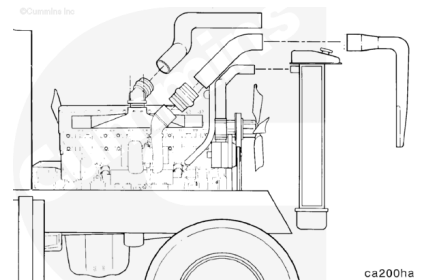
If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fittings before removing. Liquid refrigerant can cause serious eye and skin injuries.



⚠ WARNING ⚠

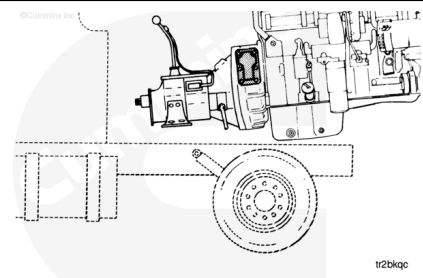
To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal law requires capturing and recycling the refrigerant.

Remove all chassis components necessary to remove the engine from the equipment.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and the transmission as an assembly without causing personal injury.



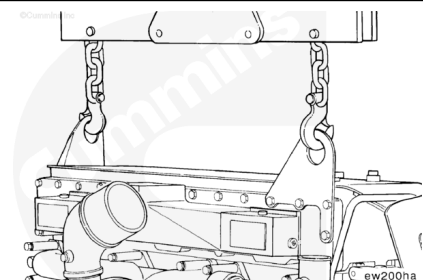
The dry weight of the engine is approximately 940 kg [2070 lb]. See the equipment manufacturer's specifications for the transmission weight.

On applications in which the rear engine mounts are attached to the transmission, it will be necessary to remove the engine and transmission as an assembly.

Use a properly rated hoist and engine lifting fixture, Part Number 3162871, attached to the engine-mounted lifting brackets, to remove the engine.

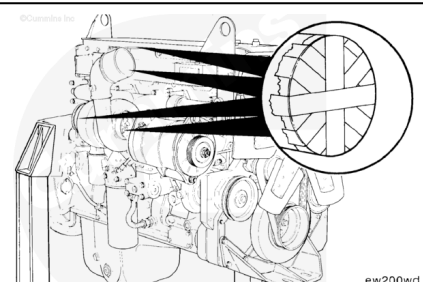
Note : If the transmission is **not** removed, place a support under the transmission to prevent it from falling.

Remove the engine.

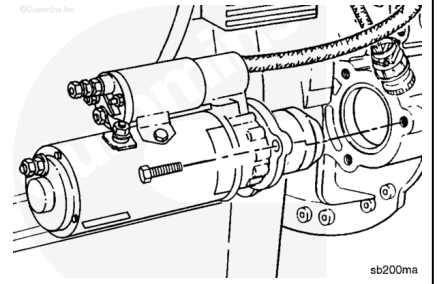


Cover all engine openings to prevent dirt and debris from entering the engine.

Place the engine on suitable engine support stands.



Remove all remaining accessories and brackets that will be used with the replacement engine.



Last Modified: 02-May-2006
